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Identification of key cardiovascular disease predictive factors from the China Health and Retirement Longitudinal Study dataset using machine learning-based algorithms

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Abstract

Background Cardiovascular diseases (CVDs) are a leading cause of death but are largely preventable. Consequently, devising effective predictive models to identify high-risk factors, and subsequently shape earlier interventional strategies, are of great importance. Machine-learning algorithms have been identified as a tool to devise such models, as they have greater flexibility and scalability versus traditional ones. This study aimed to identify predictive features for CVD risk, by applying 5 machine learning algorithms to the China Health and Retirement Longitudinal Study (CHARLS) dataset.

Methods 91,232 individuals from CHARLS, collected in 2011, 2012, 2015, and 2016, were randomly divided, in a 4:1 ratio, into training (72,985; 12,433 with CVD) and testing (18,247; 3,109 with CVD) datasets. This dataset was preprocessed by imputing missing values and standardizing features using StandardScaler, which was first fitted on the training dataset, then applied to training and testing sets, resulting in 61 features incorporated into 5 machine learning algorithms: logistic regression, random forest, eXtreme gradient boosting, light gradient-boosting machine (LightGBM), and deep neural network. The optimal hyperparameters and classification thresholds were identified for each algorithm, and their predictive capabilities assessed by receiver operating characteristic (ROC) curves. The top 15 predictive features from each algorithm were identified by Shapley Additive exPlanations (SHAP), followed by Venn diagrams and UpSet plots to determine which features were shared among all 5 algorithms.

Results Out of the 61 predictors, 9 were continuous (ex. blood pressure, biomarker levels), 6 ordinal (ex. education, self-rated health), and 46 binary (ex. smoking status, comorbidity flags). Furthermore, LightGBM had the highest

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area under the curve, 0.8002 (95% CI = 0.7917–0.8089), under ROC, indicating that it had the strongest predictive capabilities. Venn diagrams and UpSet plots found that out of the top 15 predictive features for CVD risk from each algorithm identified under SHAP, 8 were shared across all 5 algorithms: hypertension, self-rated health, dyslipidemia, age, digestive disease, retirement status, chest pain, and lung disease.

Conclusion A risk prediction model was successfully devised for CVD, involving 8 key clinically relevant and easily obtainable factors, thereby providing a useful tool for screening high-risk populations, enabling early CVD prevention and treatment.

Keywords Cardiovascular disease, China Health and Retirement Longitudinal Study, Machine learning algorithm, Shapley Additive Explanations, Risk factors

Background

Cardiovascular diseases (CVDs) are the leading causes of death worldwide, with > 19 million deaths in 2021 [1, 2]; they significantly impair patient quality of life and are associated with high risks for complications [3], leading to substantial healthcare resource utilization and societal burdens [4, 5]. However, CVDs are largely preventable [1]; consequently, developing effective predictive models to identify high-risk traits associated with them, and subsequently guide preventative intervention strategies, is of great importance. One potential tool to develop these predictive models is machine learning, which involves computational algorithms autonomously "learning" from data. Compared to traditional predictive models, machine learning possesses multiple advantages for clinical applications, such as (1) greater flexibility in their model architectures, enabling adaptive processing of un- and structured medical data, and (2) a high degree of versatility, allowing them to simultaneously accomplish multiple clinical tasks, including risk stratification, disease diagnosis, and prognosis evaluation [6]. In fact, numerous studies have shown that machine learning models generally outperform traditional statistical ones in predictive accuracy [7]. Furthermore, through applying interpretability techniques, such as Shapley Additive exPlanations (SHAP), the contribution of each variable in the machine learning model to the prediction results could be intuitively quantified, thereby enhancing model interpretability and providing transparent evidence for clinical decision-making [8, 9]. All these advantages render machine learning a powerful tool for constructing predictive models, providing methodological support for developing personalized treatments [10].

Previous studies have also noted that multiple factors, such as age [11, 12], chronic kidney disease [13], dietary factors [14], obesity [15], and metabolic diseases [16], have been associated with CVD etiology and pathogenesis, and thereby increased risk. These factors may exhibit complex, non-linear associations with CVDs [17–19], which machine learning methods could potentially capture that would otherwise be missed by traditional predictive models [20].

Based on these observations, in this study, we identified 61 predictors, associated with CVD occurrence, from middle-age-to-elderly populations in the China Health and Retirement Longitudinal Study (CHARLS) database, and incorporated them into 5 machine learning algorithms: logistic regression (LR), random forest (RF), eXtreme gradient boosting (XGBoost), light gradient-boosting machine (LightGBM), and deep neural networks (DNN); there, the most accurate was LightGBM. The top 15 predictors for each of those algorithms were identified by SHAP analysis, in which all 5 algorithms shared the following predictors: hypertension, self-rated health, dyslipidemia, age, digestive disease, retirement status, chest pain, and lung disease. These predictors were incorporated into the following website https://charls-cvd-risk-demo-2026.streamlit.app/, enabling individual visitors to easily predict their own CVD risk levels. Therefore, they provide a more accessible CVD risk assessment method for individuals and clinicians, in turn facilitating the development of early CVD interventional strategies.

Methods

Obtaining CHARLS patient data and preprocessing

This study utilized data from CHARLS, a longitudinal cohort study of the ≥ 45-year-old Chinese population [21], which focused on examining aging and health statuses for middle-aged-elderly individuals. It has extensive demographics, health conditions, socioeconomic status, and biometrics data [21]. Here, we used CHARLS data collected in 2011, 2012, 2015, and 2016, comprising a total of 96,627 individuals; those with > 20% of features (variables) missing were excluded, yielding a final dataset of 91,232 participants (Fig. 1). It is worth noting that this dataset also had 15 participants < 45-years-old that were part of the CHARLS database, even though it was aimed at ≥ 45-year-olds. Nevertheless, the final dataset was randomly divided, in a 4:1 ratio, into training (72,985; 12,433 with CVD) and testing (18,247; 3,109 with CVD) datasets [22] (Fig. 1). Identification of CVD in CHARLS was based on questionnaires from individuals asking if they had been diagnosed with heart disease by

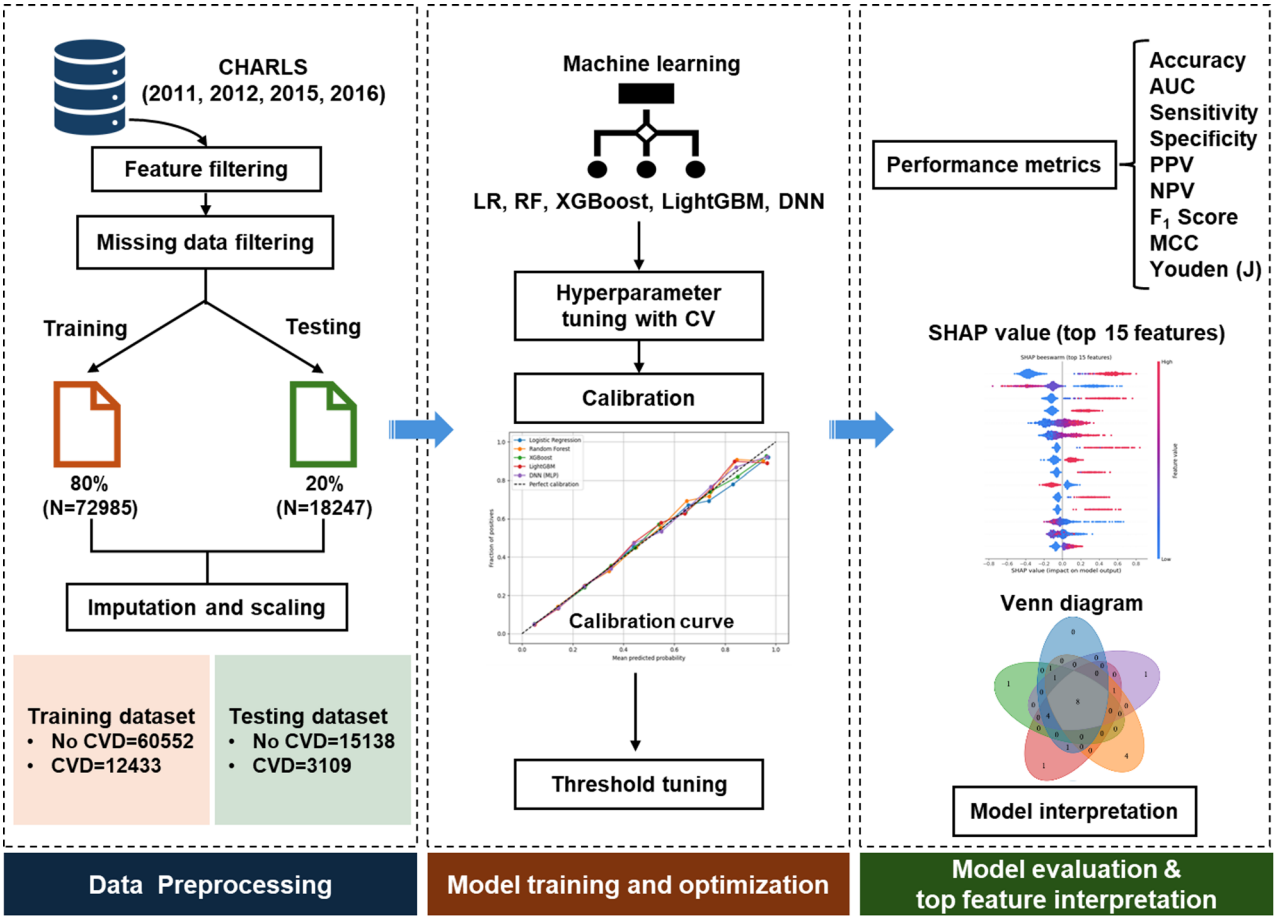


Fig. 1 Schematic overview of the complete data processing and modeling workflow for identifying predictive factors for cardiovascular disease (CVD) risk, including data splitting, imputation, normalization, and evaluation

a doctor, and was defined as an incident event, including coronary heart disease, heart attack (myocardial infarction), angina, congestive heart failure, and other general heart problems, while excluding stroke; this was based on CHARLS variable da007. Detailed descriptions of the survey design and procedures are available on the official website (http://charls.Pku.edu.cn/), and this study was conducted in accordance with the CHARLS data use agreement and ethical guidelines.

The CHARLS data was then preprocessed to ensure data quality, resulting in 61 features, of which 9 were continuous, 6 were ordinal, and 46 were binary, as shown in Supplemental Table S1. Subsequent preprocessing steps, including imputation and normalization, were independently performed, within each fold of the cross-validation pipeline, on the training set to avoid leakage. More specifically, for continuous and ordinal features, missing values present in both training and testing sets were imputed using the means for each feature, while for binary features, they were imputed based on the mode; all these values were calculated from the training set. After imputation, continuous features in the training set were standardized using StandardScaler in scikit-learn (1.7.0), where the scaler was fitted; this was then applied to transform the testing set, thereby ensuring that both datasets were on comparable scales while avoiding data leakage. Additionally, to prevent leakage against training/testing partitions, data splitting was performed at the participant level, ensuring that no overlapping of participants, and thus repetition, was present between the training and testing sets. Figure 1 presents a schematic overview of the complete data processing and modeling workflow, including data splitting, imputation, normalization, and evaluation.

Selecting the 5 machine learning models

We developed 5 models to identify features predictive for high CVD risk, based on the following widely used machine learning algorithms: LR, RF, XGBoost, LightGBM, and DNN. The high degree of effectiveness for these 5 algorithms served as the basis for their selection in this study. Furthermore, these models were based on a short-term (1-year) timeframe, in which event-level data

was used to forecast the incidence of a new CVD diagnosis from Year t to Year t + 1.

Identifying and optimizing the hyperparameters and classification thresholds

For these algorithms, the proper hyperparameters and classification thresholds needed to be identified and selected. Hyperparameter tuning was performed via 5-fold cross-validation on the training set. To ensure unbiased evaluations and prevent data leakage, all data preprocessing procedures, which were first defined on the entirety of the CHARLS dataset, were fitted exclusively on each training fold, then applied to their corresponding validation folds. Afterwards, stratified 5-fold cross-validation was applied, using scikit-learn, to ensure representative sampling and accurate performance estimation, in which the proportions of CVD individuals within each fold was maintained, so that they closely reflected that of the original training dataset [23]. Hyperparameter optimization was then performed using Optuna (4.4.0), an optimization framework known for efficient search and pruning capabilities [24]. There, each of the machine-learning algorithms underwent customized tuning processes to determine its optimal configuration, in the form of maximized areas under the curve (AUCs) in receiver operating characteristic (ROC) curve analyses for each algorithm. For LR, regularization strengths (C) and class weighting strategies were adjusted, while for RF, the number of estimators, maximum tree depth, and class weighting were optimized. Optimization for XGBoost involved parameter modification, such as for maximum depth, learning rate, number of estimators, subsample ratio, column sampling ratio, and scale_pos_weight to address class imbalances; the number of estimators, learning rate, and scale_pos_weight was also tuned for LightGBM, along with the number of leaves. As for DNN, the number and size of hidden layers, activation functions, L2 regularization parameters (alpha), and initial learning rates were optimized.

To determine the optimal decision threshold for our risk classifiers, 3 complementary metrics: Youden's index (J=sensitivity+specificity-1), Matthews Correlation Coefficient (MCC), and F_1-score, were applied, across a range of candidate cutoffs from 0.10 to 0.99, in increments of 0.01. The optimal threshold was defined as the value that maximized J under ROC curve analysis. The selection of J was owed to it emphasizing balanced classification performances, in terms of sensitivity and specificity; it is thus widely recognized as a robust criterion for cutoff selection [25].

Calibrating the machine-learning algorithms

To ensure that the probability outputs of the classifiers reflect the actual event likelihoods, isotonic regression was applied to calibrate the predicted probabilities. More specifically, each complete prediction pipeline, including all preprocessing steps and the base learner, was calibrated, using CalibratedClassifierCV in scikit-learn, with method="isotonic" and 5-fold cross-validation. There, for each fold, the algorithms were trained on 80% of the data, while predictions on the remaining 20% were used to fit an isotonic regression model, establishing monotonic mapping from raw scores to calibrated probabilities. Testing set predictions were then transformed by this ensemble of calibrators and evaluated under final performance and decision threshold analyses.

Evaluating algorithm performances and interpreting their results

To evaluate the 5 algorithms, we applied the optimal hyperparameters and classification threshold obtained in the hyperparameter tuning stage, and trained them on the pre-processed training set, followed by testing with the pre-processed testing set. The following metrics were computed to quantify their predictive performances, with their formulas given in Supplemental Table S2: accuracy, AUC, sensitivity (recall), specificity, precision/positive (PPV) and negative predictive values (NPV), as well as F_1-score, MCC, and J. These formulas involve the values of true positives (TP) and negatives (TN), along with false positives (FP) and negatives (FN). Furthermore, AUCs were calculated from the ROC curve to assess the discriminatory capabilities of the algorithms across all thresholds. Bootstrap resampling, in 1000 iterations, was then carried out to obtain 95% confidence intervals (CI) for each metric. Decision curve (DCA) and precision-recall (PR) curve analyses were carried out for all 5 algorithms to evaluate their clinical utility and ability to identify positive CVD cases, respectively.

The contributions of individual features to the predictive capabilities of the 5 algorithms were evaluated using SHAP values, which provide a principled, game-theoretic allocation of the marginal impact of each feature on the output of the algorithm, transforming the "black-box" decision process into interpretable attributions [26]. These values were obtained for every feature, within each algorithm, by first drawing a random subset of 1,000 observations from the held-out test set, which served as the explanation cohort, followed by the usage of PermutationExplainer in the Python SHAP library (0.47.2). These features were subsequently ranked based on their mean absolute SHAP values, and summary ("beeswarm") plots were generated for the top 15 predictors for each algorithm. Within these plots, each point corresponds to a SHAP value for a given feature, within each sample, positioned horizontally by its magnitude and colored blue-to-red, representing low-to-high feature values. These visualizations simultaneously conveyed both the

relative importance of each feature and the directionality of its effect on the model output. A Venn diagram and an UpSet Plot, using the plotly (0.9.0) and pandas (2.2.2) packages, were drawn to identify any overlaps among the top 15 features, across 5 classifiers, which represent common drivers of risk. The top 8 predictors that SHAP analysis identified across the 5 machine learning algorithms were then incorporated into a simplified "lite" model for prospective external validation, as well as the following website (https://charls-cvd-risk-demo-2026.streamlit.app/), using Python Streamlit.

Results

Obtaining the CHARLS dataset and features for the algorithms

From the CHARLS dataset, a total of 61 predictors were identified: 9 continuous (ex. blood pressure, biomarker levels), 6 ordinal (ex. education, self-rated health) and 46 binary (ex. smoking status, comorbidity flags) after we dropped any variable with > 20% of their values being missing (Supplemental Table S1). The resulting preprocessed data was then split into training (72,985) and testing (18,247) sets, each with CVD prevalence of 17.0%.

Optimizing the 5 machine learning algorithms

For each algorithm, their raw probability outputs were determined by plotting decile-based calibration curves, in which they were modestly mis-calibrated prior to calibration; more specifically, LR and RF tend to underestimate risk in the mid-range (0.3–0.6), while XGBoost and DNN had overconfidence (> 0.6; Fig. 2A). However, after applying isotonic regression calibration, the calibration curves for all 5 algorithms were close to the 45° diagonal, indicating a high degree of correspondence between adjusted probabilities and observed event rates across the full risk range (Fig. 2B). The most pronounced correction occurred for the DNN, whose pre-calibration probabilities at ranges ≥ 0.8 were reduced by > 0.10 on average, bringing its post-calibration deviation below 0.02 (Fig. 2A-B).

Based on these calibrated probabilities, the cutoff for each algorithm was identified by systematic threshold tuning, in order to obtain their maximum J values. These cutoffs were reported in Table 1, along with their AUCs, accuracies, sensitivities, specificities, PPV, NPV, MCC, F_1-score, and J; all of them had 95% confidence intervals estimated by bootstrapping. The optimization yielded the following calibrated-probability thresholds: 0.16 for both LR and DNN, 0.19 for RF, 0.45 for XGBoost, and 0.68 for LightGBM. At these cut-off points, each model achieved its maximal J, a metric that defines the optimal balance between sensitivity and specificity. Calibration, followed by threshold optimization, produced well-calibrated models, with balanced predictive performance for both positive and negative cases (Fig. 2).

LightGBM is the most optimal for predicting CVD

ROC curve analyses were performed for the five machine learning algorithms, which yielded similar AUC values approximating 0.8, with inter-model differences of approximately 0.025 being observed (Fig. 3A). Visual assessment of the violin plots indicated that the AUC values for all models followed an approximately normal distribution (Fig. 3B). Among those algorithms, LightGBM had the highest AUC, at 0.8002 (95% CI = 0.7917–0.8089), indicating that it had the strongest discriminatory capabilities; it also had the highest sensitivity (0.7404, 95% CI: 0.7250–0.7564) and NPV (0.9304,

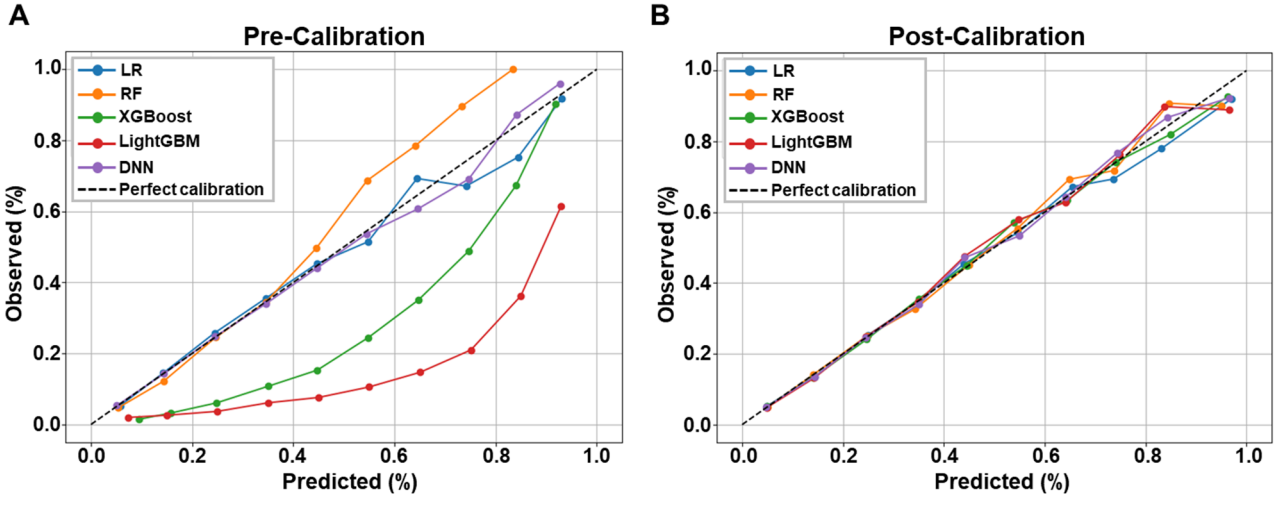


Fig. 2 Calibration curves for the 5 machine learning algorithms. **(A)** Pre- and **(B)** post-isotonic regression calibration performances on the testing dataset for logistic regression (LR), random forest (RF), eXtreme gradient boosting (XGBoost), light gradient-boosting machine (LightGBM) and deep neural networks (DNN) machine learning algorithms

Table 1 Comparing performance metrics for the 5 machine learning algorithms on the testing dataset (all with 95% confidence interval [CI])

Model	Threshold	AUC (95% CI)	Accuracy (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)	PPV (95% CI)	NPV (95% CI)	MCC (95% CI)	F_1-score (95% CI)	J (95% CI)
LR	0.16	0.7921 (0.7832–0.8004)	0.7184 (0.7115–0.7249)	0.7276 (0.7119–0.7432)	0.7165 (0.7091–0.7240)	0.3451 (0.3332–0.3569)	0.9276 (0.9230–0.9322)	0.3480 (0.3332–0.3617)	0.4681 (0.4553–0.4802)	0.4441 (0.4260–0.4609)
RF	0.19	0.7982 (0.7894–0.8065)	0.7327 (0.7263–0.7391)	0.7211 (0.7055–0.7366)	0.7351 (0.7282–0.7421)	0.3586 (0.3465–0.3707)	0.9277 (0.9230–0.9322)	0.3614 (0.3465–0.3757)	0.4790 (0.4662–0.4914)	0.4562 (0.4386–0.4728)
XGBoost	0.45	0.7924 (0.7840–0.8013)	0.7190 (0.7123–0.7254)	0.7231 (0.7068–0.7406)	0.7181 (0.7111–0.7253)	0.3451 (0.3335–0.3574)	0.9266 (0.9218–0.9316)	0.3462 (0.3320–0.3610)	0.4671 (0.4548–0.4800)	0.4412 (0.4236–0.4594)
LightGBM	0.68	0.8002 (0.7917–0.8089)	0.7172 (0.7108–0.7236)	0.7404 (0.7250–0.7564)	0.7124 (0.7052–0.7197)	0.3457 (0.3343–0.3579)	0.9304 (0.9256–0.9351)	0.3536 (0.3395–0.3684)	0.4714 (0.4588–0.4843)	0.4528 (0.4358–0.4705)
DNN	0.16	0.7921 (0.7836–0.8007)	0.7097 (0.7031–0.7162)	0.7402 (0.7241–0.7559)	0.7035 (0.6963–0.7105)	0.3390 (0.3276–0.3502)	0.9295 (0.9247–0.9343)	0.3451 (0.3310–0.3588)	0.4649 (0.4526–0.4767)	0.4436 (0.4261–0.4603)

LR, logistic regression; RF, random forest; XGBoost, eXtreme gradient boosting; LightGBM, light gradient-boosting machine; DNN, deep neural networks; AUC, area under the curve; PPV, positive predictive value; NPV, negative predictive value; MCC, Matthews correlation coefficient; J, Youden's index

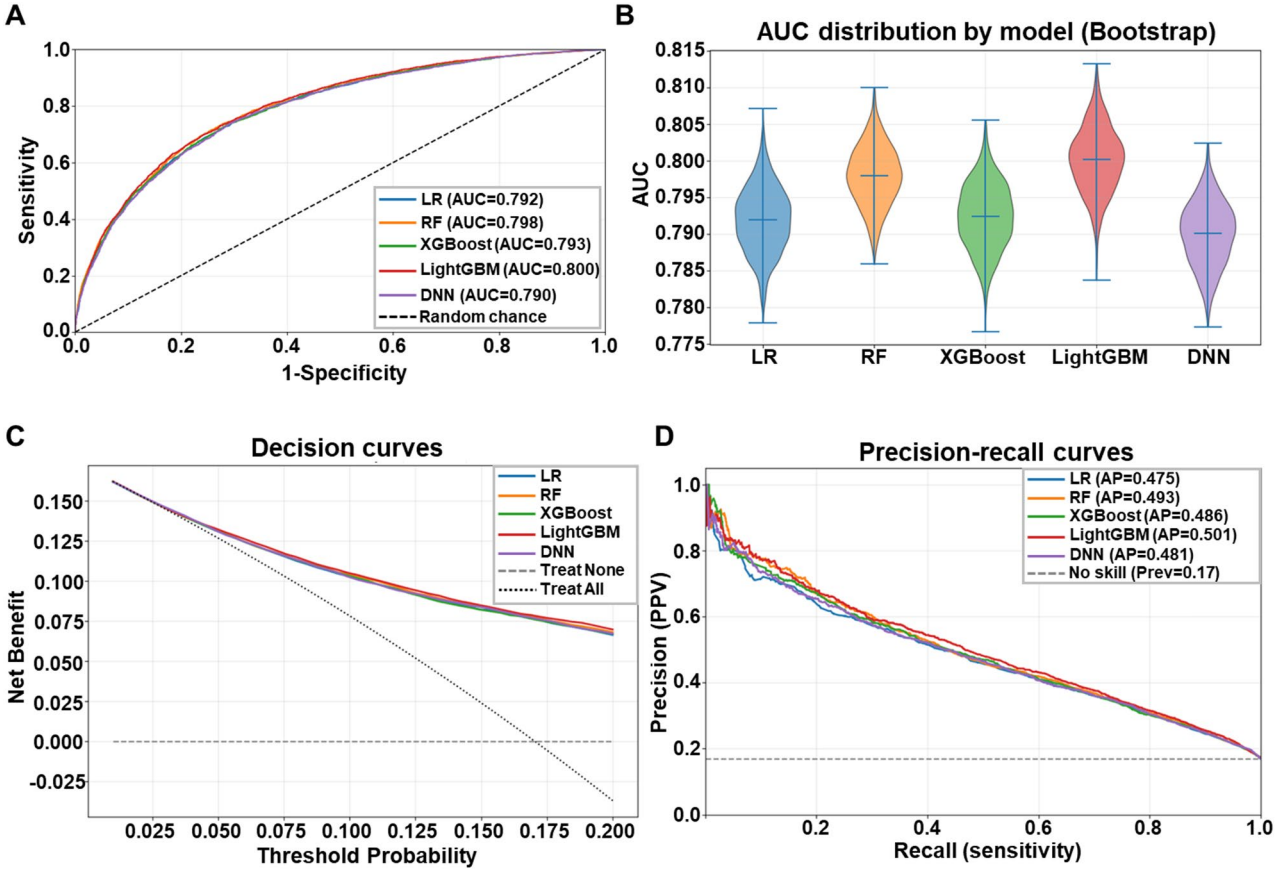


Fig. 3 Receiver operating characteristic (ROC) and decision curve analyses for the 5 machine learning algorithms. **(A)** ROC curve analyses for LR, RF, XGBoost, LightGBM, and DNN on the testing dataset. **(B)** Violin plots showing the bootstrap distributions for the areas under the curve (AUC) from the 5 machine learning algorithms on the testing dataset. Central bar indicates the medium, and whiskers represent the interquartile ranges. **(C)** Decision curve analyses for the 5 machine learning algorithms, in which those with clinical utility were identified based on them possessing areas above the intersection between "Treat None" (no treatment) and "Treat All" (all patients are treated) lines. **(D)** Precision-recall curves for the 5 machine learning algorithms, yielding average precisions (AP) for identifying positive CVD cases. Precision was expressed in terms of positive predictive values (PPV)

95% CI: 0.9256–0.9351). However, RF was the highest for accuracy (0.7327, 95% CI = 0.7263–0.7391), specificity (0.7351, 95% CI = 0.7282–0.7421), MCC (0.3614, 95% CI = 0.3465–0.3757), F_1-score (0.4790, 95% CI = 0.4662–0.4914), J (0.4562, 95% CI = 0.4386–0.4728), and PPV (0.3586, 95% CI = 0.3465–0.3707), reflecting a strong, balanced performance across multiple criteria. Therefore, among the 5 algorithms, LightGBM had the most superior discriminatory capabilities, while RF was the most consistent among various threshold-dependent metrics. Furthermore, DCA indicated that all 5 algorithms provided similar net benefits; their clinical utility was therefore greater than treating all, or none of the CVD patients (Fig. 3C). Additionally, PR curve analysis for all 5 machine learning algorithms indicated that no significant differences in their performances to identify positive cases, in terms of average precisions, were present (Fig. 3D).

Identifying the top 15 CVD predictors under SHAP, Venn, and UpSet diagrams

We then identified the top 15 predictors for each of the 5 algorithms using SHAP analysis, of which higher SHAP values for key clinical and health-related features, such as hypertension, dyslipidemia, and poor self-rated health, were associated with increases in predicted CVD risk for all 5 algorithms (Fig. 4). On the other hand, some demographic or cognitive features, such as education level, were inversely associated with higher CVD risk (Fig. 4).

Afterwards, we identified, using a Venn diagram, any overlapping predictors among all 5 algorithms, of which there were 8, ranked from most to least important in terms of SHAP: hypertension, self-rated health, dyslipidemia, age, digestive disease, retirement status, chest pain, and lung disease (Fig. 5A). These 8 were thus considered to be robust risk factors for CVD. On top of those 8, 4 additional features were uniquely present for RF, while 4 others were shared by the other 4 algorithms, but not RF, highlighting areas of consensus and divergence, and suggesting that the relative importance of certain predictors may be algorithm-dependent (Fig. 5B). Notably, RF exhibited the greatest divergence, sharing fewer top predictors with the other models (Fig. 5B).

The top 8 predictors, identified by SHAP analysis, across the 5 machine learning algorithms were then incorporated into a simplified "lite" model for prospective external validation; their selection was to obtain a consensus-based feature set for prospective validation.

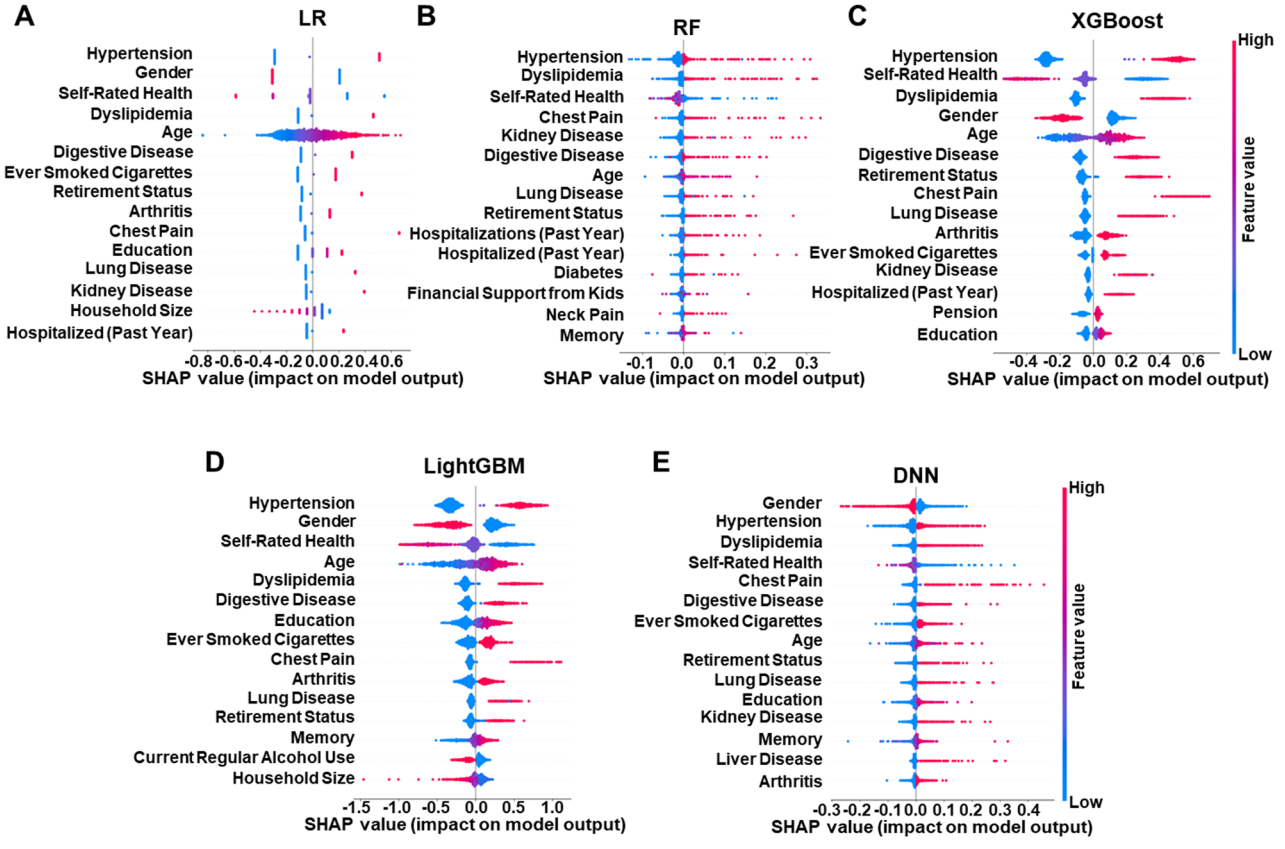


Fig. 4 Shapley additive exPlanations (SHAP) summary plots for the top 15 predictive features of the 5 machine learning algorithms. SHAP plots for **(A)** LR, **(B)** RF, **(C)** XGBoost, **(D)** LightGBM, and **(E)** DNN. Features were ranked from most to least important. Positive SHAP indicated increased predicted CVD risk, while negative was decreased risk. Values for each individual feature were indicated from low (blue) to high (red)

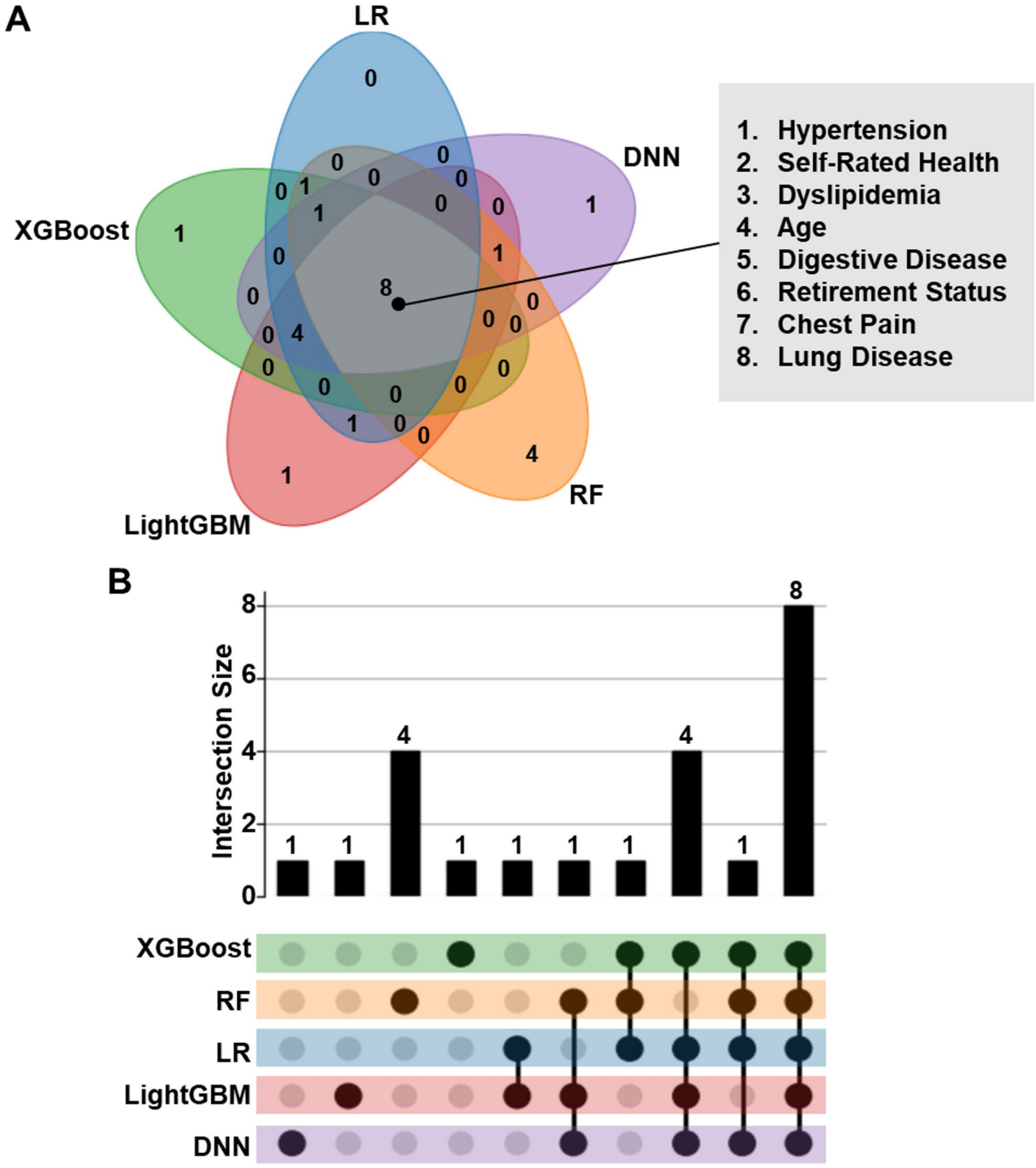


Fig. 5 Eight predictive features, out of the top 15 identified by SHAP, were shared by all 5 machine learning algorithms. **(A)** Venn diagram indicating the shared features, ranked from most to least important in terms of SHAP, among LR, RF, XGBoost, LightGBM, and DNN. **(B)** UpSet plot showing the number of shared and unique features among the 5 machine learning algorithms. Bar graphs indicated the number of features present among the different algorithm combinations, indicated by the matrix below

This model, which served as the basis for the following website, comprising of a web-based survey and calculator: https://charls-cvd-risk-demo-2026.streamlit.app/, yielded an AUC of 0.7662 under ROC curve analysis with k-fold cross-validation, compared to the AUC of 0.80 obtained for the original risk prediction model. Owing to the "lite" model retaining > 95% of the predictive performance of the original model, but lowering the feature

space by 87%, the risk of overfitting was reduced. This "lite" model therefore enables any individual to easily predict their own CVD risk levels, facilitating prospective external validation analyses for clinical readiness.

Discussion

This study was based on the CHARLS database and employed 5 machine learning algorithms to construct predictive models for CVDs, along with systematically screening for relevant risk factors. The following feature categories were incorporated: (1) Personal history, including age, gender, smoking and alcohol consumption history, etc., (2) Social interaction, such as community activity participation and social interaction frequencies, (3) Quality of life, namely the capabilities to conduct daily activities like eating, dressing, walking, and breathing, (4) Comorbidities, including diabetes, hypertension, dyslipidemia digestive system, chronic kidney, and respiratory diseases, etc., and (5) Mental health status, in the form of mental state assessments and depression rating scales. Model interpretability was assessed using SHAP analysis, where by consolidating the top contributors from each of the 5 algorithms, a consensus set of 8 core features were identified as being the most stable and significant CVD risk predictors: hypertension, self-rated health status, dyslipidemia, age, retirement status, chest pain symptoms, plus digestive system and respiratory diseases. SHAP analyses were also able to elucidate the contribution of each of these features for individualized risk assessments, providing quantitative evidence for clinical risk stratification. The top 8 predictors identified by SHAP analysis were incorporated into a simplified "lite" model, which served as the basis for the following website https://charls-cvd-risk-demo-2026.streamlit.app/, enabling individual visitors to easily predict their own CVD risk levels. Overall, the leveraging multi-model ensemble analyses and artificial intelligence-based algorithms offered novel methodological insights for developing personalized early detection and interventional approaches for CVD.

CVD prevention is a key strategy for reducing its disease burden [1, 27], and accurately identifying individual risk is crucial for early intervention. Multiple predictive models have already been developed to evaluate CVD risk, incorporating traditional risk factors, such as systolic blood pressure and age, that have been noted to play significant roles [28]. Furthermore, a UK Biobank-based study identified the influence of 20 additional features, including hypertension and self-reported health ratings, in predictive CVD models [29], which are in line with Wan et al. identifying an association between dyslipidemia and CVD [30]. Additionally, recent research has revealed that digestive system diseases, such as inflammatory bowel and non-alcoholic fatty liver diseases, are significantly associated with increased CVD risk [31, 32], which could be due to endothelial dysfunction, chronic inflammatory responses, and abnormal lipid metabolism [32]. Strong links have also been documented between pulmonary diseases, such as chronic obstructive pulmonary disease, lung cancer, and pulmonary hypertension, with increased CVD risk [33–35]; this was the case even for mild lung functional impairments [36]. Therefore, numerous risk factors and co-morbidities have been linked to CVD occurrence; in particular, our identification of hypertension, self-rated health, dyslipidemia, age, plus digestive and lung diseases were in line with the previous findings.

However, for 2 other factors, namely retirement status and chest pain, their association with CVD risk in our study needs to be treated with caution. This is owed to the mixed picture present in the current literature, where even though a number of studies linked retirement with CVD risk [37, 38], Moon et al. argued that this association is not likely due to retirement itself contributing to CVD, but rather the influence of the health status of an individual on their decision to retire [38]. Likewise, for chest pain, Rovai et al. pointed out that only relying on chest pain as a predictive indicator may be of limited utility [39], though some studies also observed that chest pain patients, versus those without, had significantly higher CVD risk [40, 41]. Based on these observations, the veracity of these 2 factors for influencing CVD risk should be further validated.

In light of multiple factors being found to be linked to CVD occurrence and progression, their comprehensive assessment could potentially significantly improve predictive accuracies for the disease. For instance, existing studies have shown that constructing multivariate risk models is a key step for epidemiological research on CVDs [42–44]. In line with these studies, we carefully selected 8 key predictive indicators, falling into 3 different categories: underlying disease conditions (hypertension, digestive system diseases, dyslipidemia, pulmonary diseases), demographic characteristics (age, retirement status, self-rated health), and typical symptoms (chest pain), striking a balance between clinical practicality and predictive efficacy. The resulting multi-dimensional assessment system not only enhances its predictive power [26], but also ensures its feasibility in clinical applications. This is due to the indicators being easily obtainable in routine clinical practice, as well as only 8 being present, which is in line with < 10 features being used in other predictive models [44, 45], all of which enables the predictive model to maintain its accuracy, while avoiding excessive complexity, thus rendering it practical for clinical decision-making.

Currently, there are a number of traditional risk prediction models for CVD, such as the American College of

Cardiology (ACC)/American Heart Association (AHA) risk calculator. This model, though, has been noted to overestimate CVD risk in different populations [46, 47], and its predictive accuracy is based on a 10-year, or lifetime span [48], rather than the 1-year period for our machine learning-based risk prediction model. Consequently, traditional models like the ACC/AHA risk calculator may not be as useful for predicting shorter-term CVD incidents. Furthermore, machine learning-based predictive models, on top of lowering the risk for false positives and enabling shorter-term predictions, have been documented by Huang et al. to be more accurate than traditional models [49]. Overall, machine learning-based models have advantages over traditional models in terms of accuracy, timespan flexibility, and ability to handle non-linear factors, for predicting CVD occurrence. Our analysis of large-scale populations thus allowed us to identify CVD risk factors with machine learning and establish a prediction model, which could supplement traditional prediction models to improve CVD predictive specificity and sensitivity. We incorporated it as part of a website (https://charls-cvd-risk-demo-2026.streamlit.app/), which could facilitate simpler, more convenient screening of high-risk populations.

There are a number of limitations to this study, one of which is our usage of mean/mode imputation for missing values in data preprocessing. Mean/mode imputation has been previously noted to potentially distort distributions and underestimate uncertainty, owing to imputed values being treated as though they were observed ones [50, 51]. As a result, sensitivity analyses (ex. multiple imputations) have been regarded to provide more support for inferences; indeed, that has been observed in previous studies [50, 51]. However, this study is primarily a predictive, risk-stratification study, rather than an inferential analysis aimed at unbiased estimation of regression coefficients and their standard errors. In predictive studies, multiple imputations may not always yield superior performances to that of mean/mode imputation, as missing-data recommendations, developed for inferences, may not directly translate [52, 53]. Future studies should aim to utilize multiple imputations in sensitivity analysis for dealing with missing values in datasets. Another limitation is that we did not include continuous scale characteristics, particularly systolic blood pressure and lipid profiles, in our predictive model. However, this exclusion was based on the application of a strict quality threshold during pre-processing, in which any predictor with a "missing-ness" rate > 20% was removed. Indeed, systolic blood pressure, total cholesterol, and low-density lipoprotein cholesterol (LDL) were among those predictors exceeding that rate. Furthermore, this machine learning-based predictive model, unlike with traditional models designed for healthcare providers, was oriented for user-led self-checks and primary prevention. As a result, risk factors requiring clinical laboratory values, such as for LDL, or blood pressure readings would create a high barrier to entry for individuals. Therefore, by focusing on more accessible markers in the 61 identified predictors, such as diagnosis codes and symptoms, the resulting model was developed as a simple, practical tool for individuals to assess their risk, without requiring access to clinical equipment or lab results.

Conclusions

This study successfully constructed a risk prediction model for CVD, and identified the hierarchical importance of key predictive factors, such as hypertension, dyslipidemia, and age, via SHAP analysis. In particular, hypertension, self-rated health status, dyslipidemia, age, digestive system diseases, retirement status, chest pain symptoms, and pulmonary diseases were found to be the core factors associated with CVD risk. This prediction model, which was incorporated into this website: https://charls-cvd-risk-demo-2026.streamlit.app/, therefore provides a novel decision-making tool for screening high-risk populations, in turn facilitating early CVD prevention and interventions. Future studies should confirm the predictive efficacy of these factors in larger sample sizes, at multiple centers, and with external validation, to advance their translation into clinical practice.

Abbreviations

AUC	Area under the curve
CHARLS	China Health and Retirement Longitudinal Study
CI	Confidence intervals
CVD	Cardiovascular disease
DNN	Deep neural networks
FN	False negative
FP	False positive
J	Youden's index
LightGBM	Light gradient-boosting machine
LR	Logistic regression
MCC	Matthews Correlation Coefficient
NPV	Negative predictive value
PPV	Positive predictive value
RF	Random forest
ROC	Receiver operating characteristic
SHAP	Shapley Additive exPlanations
TN	True negative
TP	True positive
XGBoost	eXtreme gradient boosting

Supplementary Information

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Supplementary Material 1

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Author contributions
JYC conceptualization, data curation, formal analysis, investigation, project administration, visualization, writing-original draft; CF data curation, formal analysis, investigation, project administration, visualization, writing-original draft; YW data curation, formal analysis, investigation, writing-original draft; YL methodology, formal analysis, writing-original draft; HCJ methodology, formal analysis, writing-original draft; TYG methodology, formal analysis, investigation, visualization, writing-original draft; CTT methodology, writing-original draft; DXH writing-review & editing; CFC writing-review & editing; PM writing-review & editing; CFL resources, writing-review & editing; WCO resources, project administration, funding acquisition, writing-review & editing; SML conceptualization, funding acquisition, supervision, writing-review & editing; YZ conceptualization, resources, software, supervision, validation, writing-review & editing; CYZ conceptualization, funding acquisition, supervision, writing-review & editing.

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Data availability
The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
This study was conducted in accordance with the China Health and Retirement Longitudinal Study data use agreement and ethical guidelines. CHARLS data was approved by the Ethics Committee of Peking University. Ethical approval numbers are IRB00001052-11015 for the main household survey, and IRB00001052-11014 for biomarker collection. This study was in compliance with the Declaration of Helsinki. All participants provided written informed consent.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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References

1. Kaminsky LA, German C, Imboden M, Ozemek C, Peterman JE, Brubaker PH. The importance of healthy lifestyle behaviors in the prevention of cardiovascular disease. Prog Cardiovasc Dis. 2022;70:8–15.
2. Martin SS, Aday AW, Allen NB, Almarzooq ZI, Anderson CAM, Arora P, Avery CL, Baker-Smith CM, Bansal N, Beaton AZ, et al. 2025 heart disease and stroke statistics: A report of US and global data from the American heart association. Circulation. 2025;151:e41–660.
3. Jha MK, Qamar A, Vaduganathan M, Charney DS, Murrough JW. Screening and management of depression in patients with cardiovascular disease: JACC state-of-the-art review. J Am Coll Cardiol. 2019;73:1827–45.
4. Dunlay SM, Roger VL, Redfield MM. Epidemiology of heart failure with preserved ejection fraction. Nat Rev Cardiol. 2017;14:591–602.
5. Steven S, Frenis K, Oelze M, Kalinovic S, Kuntic M, Bayo Jimenez MT, Vujacic-Mirski K, Helmstadter J, Kroller-Schon S, Munzel T, Daiber A. Vascular inflammation and oxidative stress: major triggers for cardiovascular disease. Oxid Med Cell Longev. 2019;2019:7092151.
6. Ngiam KY, Khor IW. Big data and machine learning algorithms for health-care delivery. Lancet Oncol. 2019;20:e262–73.
7. Zhao X, Lu Y, Li S, Guo F, Xue H, Jiang L, Wang Z, Zhang C, Xie W, Zhu F. Predicting renal function recovery and short-term reversibility among acute kidney injury patients in the ICU: comparison of machine learning methods and conventional regression. Ren Fail. 2022;44:1326–37.
8. Guan C, Gong A, Zhao Y, Yin C, Geng L, Liu L, Yang X, Lu J, Xiao B. Interpretable machine learning model for new-onset atrial fibrillation prediction in critically ill patients: a multi-center study. Crit Care. 2024;28:349.
9. Lundberg SM, Erion G, Chen H, DeGrave A, Prutkin JM, Nair B, Katz R, Himmelfarb J, Bansal N, Lee SI. From local explanations to global Understanding with explainable AI for trees. Nat Mach Intell. 2020;2:56–67.
10. Deo RC. Machine learning in medicine. Circulation. 2015;132:1920–30.
11. North BJ, Sinclair DA. The intersection between aging and cardiovascular disease. Circ Res. 2012;110:1097–108.
12. Costantino S, Paneni F, Cosentino F. Ageing, metabolism and cardiovascular disease. J Physiol. 2016;594:2061–73.
13. Baaten C, Vondenhoff S, Noels H. Endothelial cell dysfunction and increased cardiovascular risk in patients with chronic kidney disease. Circ Res. 2023;132:970–92.
14. Dynka D, Kowalcze K, Charuta A, Paziewska A. The ketogenic diet and cardiovascular diseases. Nutrients. 2023;15.
15. Powell-Wiley TM, Poirier P, Burke LE, Despres JP, Gordon-Larsen P, Lavie CJ, Lear SA, Ndumele CE, Neeland IJ, Sanders P, et al. Obesity and cardiovascular disease: A scientific statement from the American heart association. Circulation. 2021;143:e984–1010.
16. Hill MA, Yang Y, Zhang L, Sun Z, Jia G, Parrish AR, Sowers JR. Insulin resistance, cardiovascular stiffening and cardiovascular disease. Metabolism. 2021;119:154766.
17. Cribb L, Sha R, Yiallourou S, Grima NA, Cavuoto M, Baril AA, Pase MP. Sleep regularity and mortality: a prospective analysis in the UK biobank. Elife. 2023;12.
18. Gao Y, Wang M, Wang R, Jiang J, Hu Y, Wang W, Wang Y, Li H. The predictive value of the hs-CRP/HDL-C ratio, an inflammation-lipid composite marker, for cardiovascular disease in middle-aged and elderly people: evidence from a large National cohort study. Lipids Health Dis. 2024;23:66.
19. Jo U, Park K. Carbohydrate intake and risk of cardiovascular disease: a systematic review and meta-analysis of prospective studies. Nutrients. 2023;15.
20. Akbilgic O, Davis RL. The promise of machine learning: when will it be delivered? J Card Fail. 2019;25:484–5.
21. Zhao Y, Hu Y, Smith JP, Strauss J, Yang G. Cohort profile: the China health and retirement longitudinal study (CHARLS). Int J Epidemiol. 2014;43:61–8.
22. Wahl S, Boulesteix AL, Zierer A, Thorand B, van de Wiel MA. Assessment of predictive performance in incomplete data by combining internal validation and multiple imputation. BMC Med Res Methodol. 2016;16:144.
23. Szeghalmy S, Fazekas A. A comparative study of the use of stratified cross-validation and distribution-balanced stratified cross-validation in imbalanced learning. Sens (Basel). 2023;23.
24. Shao H, Liu X, Zong D, Song Q. Optimization of diabetes prediction methods based on combinatorial balancing algorithm. Nutr Diabetes. 2024;14:63.
25. Hughes G. Youden's index and the weight of evidence. Methods Inf Med. 2015;54:198–9.
26. Qi X, Wang S, Fang C, Jia J, Lin L, Yuan T. Machine learning and SHAP value interpretation for predicting comorbidity of cardiovascular disease and cancer with dietary antioxidants. Redox Biol. 2025;79:103470.
27. McConnachie A, Walker A, Robertson M, Marchbank L, Peacock J, Packard CJ, Cobbe SM, Ford I. Long-term impact on healthcare resource utilization of Statin treatment, and its cost effectiveness in the primary prevention of cardiovascular disease: a record linkage study. Eur Heart J. 2014;35:290–8.

28. Cai R, Wu X, Li C, Chao J. Prediction models for cardiovascular disease risk in the hypertensive population: a systematic review. J Hypertens. 2020;38:1632–9.
29. Alaa AM, Bolton T, Di Angelantonio E, Rudd JHF, van der Schaar M. Cardiovascular disease risk prediction using automated machine learning: A prospective study of 423,604 UK biobank participants. PLoS ONE. 2019;14:e0213653.
30. Wan H, Wu H, Wei Y, Wang S, Ji Y. Novel lipid profiles and atherosclerotic cardiovascular disease risk: insights from a latent profile analysis. Lipids Health Dis. 2025;24:71.
31. Chen B, Collen LV, Mowat C, Isaacs KL, Singh S, Kane SV, Farraye FA, Snapper S, Jneid H, Lavie CJ, Krittanawong C. Inflammatory bowel disease and cardiovascular diseases. Am J Med. 2022;135:1453–60.
32. Duell PB, Welty FK, Miller M, Chait A, Hammond G, Ahmad Z, Cohen DE, Horton JD, Pressman GS, Toth PP, et al. Nonalcoholic fatty liver disease and cardiovascular risk: A scientific statement from the American heart association. Arterioscler Thromb Vasc Biol. 2022;42:e168–85.
33. Zhang S, Liu L, Shi S, He H, Shen Q, Wang H, Qin S, Chang J, Zhong R. Bidirectional association between cardiovascular disease and lung cancer in a prospective cohort study. J Thorac Oncol. 2024;19:80–93.
34. Andre S, Conde B, Fragoso E, Boleo-Tome JP, Areias V, Cardoso J. Cronica GD-GdInDPO: COPD and cardiovascular disease. Pulmonology. 2019;25:168–76.
35. Rosenkranz S, Gibbs JS, Wachter R, De Marco T, Vonk-Noordegraaf A, Vachiery JL. Left ventricular heart failure and pulmonary hypertension. Eur Heart J. 2016;37:942–54.
36. Ramalho SHR, Shah AM. Lung function and cardiovascular disease: A link. Trends Cardiovasc Med. 2021;31:93–8.
37. Tezuka K, Kubota Y, Ohira T, Muraki I, Hayama-Terada M, Shimizu Y, Imano H, Shirai K, Okada T, Kiyama M, Iso H. Retirement status after the age of 60 years modifies the association between anger expression and the risk of cardiovascular disease: the circulatory risk in communities study. Geriatr Gerontol Int. 2024;24:385–9.
38. Moon JR, Glymour MM, Subramanian SV, Avendano M, Kawachi I. Transition to retirement and risk of cardiovascular disease: prospective analysis of the US health and retirement study. Soc Sci Med. 2012;75:526–30.
39. Rovai D, Neglia D, Lorenzoni V, Caselli C, Knuuti J, Underwood SR, Investigators ES. Limitations of chest pain categorization models to predict coronary artery disease. Am J Cardiol. 2015;116:504–7.
40. Jordan KP, Rathod-Mistry T, van der Windt DA, Bailey J, Chen Y, Clarson L, Denaxas S, Hayward RA, Hemingway H, Kyriacou T, Mamas MA. Determining cardiovascular risk in patients with unattributed chest pain in UK primary care: an electronic health record study. Eur J Prev Cardiol. 2023;30:1151–61.
41. Croft PR, Thomas E. Chest pain and subsequent consultation for coronary heart disease: a prospective cohort study. Br J Gen Pract. 2007;57:40–4.
42. Segan L, Canovas R, Nanayakkara S, Chieng D, Prabhu S, Voskoboinik A, Sugumar H, Ling LH, Lee G, Morton J, et al. New-onset atrial fibrillation prediction: the HARMS2-AF risk score. Eur Heart J. 2023;44:3443–52.
43. Dong B, Chen Y, Yang X, Chen Z, Zhang H, Gao Y, Zhao E, Zhang C. Estimated glucose disposal rate outperforms other insulin resistance surrogates in predicting incident cardiovascular diseases in cardiovascular-kidney-metabolic syndrome stages 0–3 and the development of a machine learning prediction model: a nationwide prospective cohort study. Cardiovasc Diabetol. 2025;24:163.
44. Zhuang XD, Tian T, Liao LZ, Dong YH, Zhou HJ, Zhang SZ, Chen WY, Du ZM, Wang XQ, Liao XX. Deep phenotyping and prediction of Long-term cardiovascular disease: optimized by machine learning. Can J Cardiol. 2022;38:774–82.
45. Dimopoulos AC, Nikolaidou M, Caballero FF, Engchuan W, Sanchez-Niubo A, Arndt H, Ayuso-Mateos JL, Haro JM, Chatterji S, Georgousopoulou EN, et al. Machine learning methodologies versus cardiovascular risk scores, in predicting disease risk. BMC Med Res Methodol. 2018;18:179.
46. Muntner P, Colantonio LD, Cushman M, Goff DC Jr., Howard G, Howard VJ, Kissela B, Levitan EB, Lloyd-Jones DM, Safford MM. Validation of the atherosclerotic cardiovascular disease pooled cohort risk equations. JAMA. 2014;311:1406–15.
47. Kavousi M, Leening MJ, Nanchen D, Greenland P, Graham IM, Steyerberg EW, Ikram MA, Stricker BH, Hofman A, Franco OH. Comparison of application of the ACC/AHA guidelines, adult treatment panel III guidelines, and European society of cardiology guidelines for cardiovascular disease prevention in a European cohort. JAMA. 2014;311:1416–23.
48. Kakadiaris IA, Vrigkas M, Yen AA, Kuznetsova T, Budoff M, Naghavi M. Machine learning outperforms ACC / AHA CVD risk calculator in MESA. J Am Heart Assoc. 2018;7:e009476.
49. Huang Q, Zou X, Lian Z, Zhou X, Han X, Luo Y, Chen S, Wang Y, Wu S, Ji L. Predicting cardiovascular outcomes in Chinese patients with type 2 diabetes by combining risk factor trajectories and machine learning algorithm: a cohort study. Cardiovasc Diabetol. 2025;24:61.
50. White IR, Royston P, Wood AM. Multiple imputation using chained equations: issues and guidance for practice. Stat Med. 2011;30:377–99.
51. Li P, Stuart EA, Allison DB. Multiple imputation: A flexible tool for handling missing data. JAMA. 2015;314:1966–7.
52. Sperrin M, Martin GP, Sisk R, Peek N. Missing data should be handled differently for prediction than for description or causal explanation. J Clin Epidemiol. 2020;125:183–7.
53. Sisk R, Sperrin M, Peek N, van Smeden M, Martin GP. Imputation and missing indicators for handling missing data in the development and deployment of clinical prediction models: A simulation study. Stat Methods Med Res. 2023;32:1461–77.

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